

PROBLEMS CLASS 1

1. A simple epidemic model

Consider a population which is either susceptible (S) or infected (I) with a disease.

- (a) We assume that the total population, N , is constant. Given the population is either susceptible or infected, write down a simple equation for N .
- (b) Infected people infect susceptible people by interacting with them. By considering how we represented interactions between greenflies and ladybirds, write down an equation for dI/dt in terms of I and S .
- (c) By substituting in N , eliminate S from this equation and show that

$$\frac{dI}{dt} = \beta I(N - I). \quad (1)$$

Do you recognise this type of model from chapter 1?

- (d) Find the equilibria of this model for I , establish when they are permissible, and assess their stability. You can assess the stability either graphically, or by doing an algebraic linear stability analysis (with ε etc.)
- (e) What does this mean for the spread of the disease?

2. A slightly less simple epidemic model

More realistic is to keep what we have so far, but also to allow a proportion of the infected population to recover, after which they become susceptible again. We continue to assume that N is constant, so the same equation for N from question 1 applies.

- (a) Write down a new equation for dI/dt in terms of I and S . Once again substitute in N to remove S from this equation, and show that

$$\frac{dI}{dt} = I[\beta(N - I) - \gamma]. \quad (2)$$

- (b) Find the equilibria of this model for I , establish when they are permissible, and assess their stability. Show that the disease spreads if $R_0 > 1$, where R_0 is a ratio of terms you will find.
- (c) By considering the units of the terms in the definition of R_0 , convince yourself that it is dimensionless.

Congratulations! You have found our first definition of the famous R_0 . This term is known as the basic reproductive ratio, and as we heard plenty of in the news during the peak of the pandemic crisis, can be interpreted as the average number of susceptible people that an infected person infects in a disease-free population of size N .

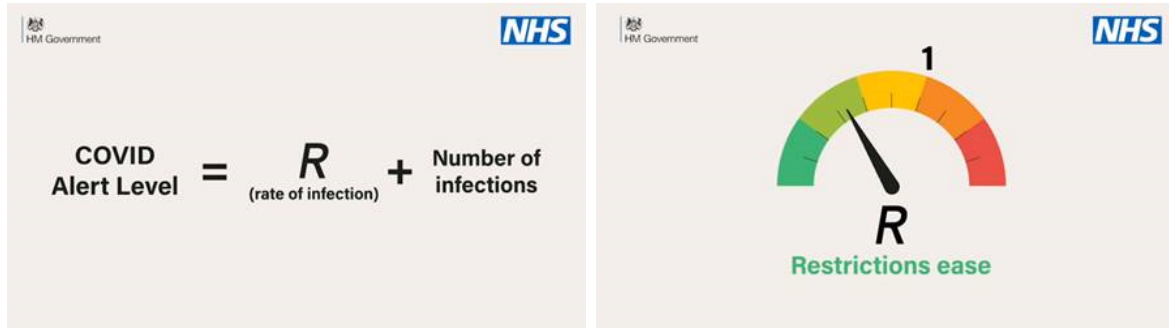


Figure 1: Government slides from coronavirus briefing, 10 May 2020

3. Quick sanity check

Take a look at the government slides from May 2020 in fig. 1, where R_0 is represented by R . Given what we have just learnt about R_0 , make a quick critique of these slides.

4. The SIR model

We now move to the popular SIR disease model. This model separates the recovered population into a new group, R , assuming that they have become immune and therefore are no longer susceptible. The total population remains constant.

- Write down equations for dS/dt , dI/dt , dR/dt and N .
- So far, we haven't added birth and natural (non-epidemic) death to our model (N has been constant). In what sort of situation might this be a reasonable choice?
- Given your equation for N , confirm that in your model, the total population is constant (i.e. that $dN/dt = 0$).
- Find all equilibria of the system. Is it/are they asymptotically stable? (Hint: pay close attention to our three possible cases at the end of section 2.2.)
- By combining two of your equations, find S as a function of R and hence demonstrate that this model predicts some individuals will always avoid infection if $S(0) > 0$.
- Solutions for the infected population, I , either show monotonic decay or rise and have a peak infection value before decaying to the equilibrium $I = 0$. Demonstrate that the criteria for the disease reaching a peak, i.e. that I ever rises, is that

$$R_0 > \frac{N}{S(0)}. \quad (3)$$